Problema 1

$A + 2B \longrightarrow C$ catalizada con Fe

$r_0 \quad [A]_0 \quad [B]_0$

$$r = k(T)\,[A]^{\alpha}\,[B]^{\beta}$$

$1+3 \longrightarrow$ $r_1 = k(T)\,[A]_1^{\alpha}\,[B]_1^{\beta}$, $r_3 = k(T)\,[A]_3^{\alpha}\,[B]_3^{\beta}$ $\Big|$ $\dfrac{r_1}{r_3} = \dfrac{\cancel{k(T)}\,\cancel{[A]_1^{\alpha}}\,[B]_1^{\beta}}{\cancel{k(T)}\,\cancel{[A]_3^{\alpha}}\,[B]_3^{\beta}} \Rightarrow \dfrac{r_1}{r_3} = \left(\dfrac{[B]_1}{[B]_3}\right)^{\beta} \Rightarrow \beta = 1$

$2+3 \longrightarrow$ $r_2 = k(T)\,[A]_2^{\alpha}\,[B]_2^{\beta}$, $r_3 = k(T)\,[A]_3^{\alpha}\,[B]_3^{\beta}$ $\Big|$ $\dfrac{r_2}{r_3} = \dfrac{\cancel{k(T)}\,[A]_2^{\alpha}\,\cancel{[B]_2^{\beta}}}{\cancel{k(T)}\,[A]_3^{\alpha}\,\cancel{[B]_3^{\beta}}} \Rightarrow \dfrac{r_2}{r_3} = \left(\dfrac{[A]_2}{[A]_3}\right)^{\alpha} \Rightarrow \alpha = 2$

$4 \rightarrow r_4 = k(T)\,[A]_4^{\alpha}\,[B]_4^{\beta} \Rightarrow k(T) = \dfrac{r_4}{[A]_4^{\alpha}\,[B]_4^{\beta}}$ $[M^{-1}s^{-1}]$. $\dfrac{Ms^{-1}}{M^{\alpha}M^{\beta}} = \dfrac{Ms^{-1}}{M^3}$

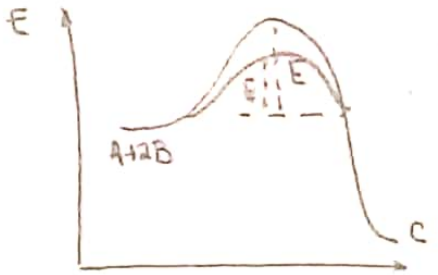
$r(B) = -2\,r(C)$

$\alpha A + \beta B \rightarrow \gamma C + \delta D \Rightarrow \dfrac{1}{\alpha}\dfrac{d[A]}{dt} = \dfrac{1}{\beta}\dfrac{d[B]}{dt} = \dfrac{1}{\gamma}\dfrac{d[C]}{dt} = \dfrac{1}{\delta}\dfrac{d[D]}{dt}$.

$A + 2B \rightarrow C \Rightarrow \underbrace{\dfrac{1}{-1}\dfrac{d[A]}{dt}}_{r(A)} = \underbrace{\dfrac{1}{-2}\dfrac{d[B]}{dt}}_{r(B)} = \underbrace{\dfrac{1}{1}\dfrac{d[C]}{dt}}_{r(C)}$.

$\alpha = 1 \quad \beta = 2 \quad \gamma = 1$

- $r(B) = -2\,r(C)$ ✓
- $r = r(A)$ ✗
- $r(A) = 2\,r(B)$



$E_{cat} < E$

PROBLEMA 4

$m = 2{,}32\,g$ $\quad XH_4$ (gas) $\quad$ átomo $X = 1{,}99 \times 10^{-23}\,g$ $\quad \rho_{N_2} = 1{,}25\,g/L$

$V = 3{,}25\,L$

1 molécula $XH_4 \rightarrow$ 1 mol at $X \rightarrow 1 \times 6{,}02 \times 10^{23}$ at. x.

1 mol at $X \rightarrow 6{,}02 \times 10^{23}$ at $X \longrightarrow x = 12\,g = C$.

1 at $X \longrightarrow 1{,}99 \times 10^{-23}\,g$

Problema 7

cuando hay N_2 → $\rho_{N_2} = 1{,}25 \frac{g}{L} = \frac{masa}{V}$

$V = 3{,}25 L$ ⇒ $masa = 1{,}25 \frac{g}{L} \cdot 3{,}25 L$ ⇒ masa de $N_2 = 4g$

moles de $N_2 = \frac{4g}{28g/mol} = 0{,}14 mol.$

CHa ⇒ $0{,}14 mol$ → $2{,}32g$

$1 mol$ → $x = 16{,}24g$ ⇒ $C + a \cdot H = X$

$12g + a \cdot 1g = 16{,}24g$

$a \cdot 1g = 4{,}24g$

$\boxed{a = 4}$

Problema 7

Butanol ($C_4H_{10}O$) → $\rho_{but} = 0{,}81$ g/mL

10 mL agua → $\rho_w = 1$ g/mL

\+ 4 mL butanol ⇒ $4 \cdot 0{,}81$ g = 3,24 g.

⇓

butanol m = ? [g]
butanol + agua 7,71 % m/m

but agua
sto + sve

7,71 % m/m = 7,71 g butanol → 100 g SL

7,71 g butanol → 92,3 g agua

0,835 g but = x ← 10 g agua

3,24 g − 0,84 g = 2,4 g. butanol sin disolver.
total but + agua.

Problema 8

25 g O_2(g) → T = 117°C = 390 K, P = 0,25 atm. + N_2(g) ⇌ 2 O_2(g) $\rho = [g/L]$.

PV = nRT ⇒ $n_T = n_{O_2} + n_{N_2}$

$PM \cdot O_2 = 32$ g/mol

25 g O_2 → $n_{O_2} = 0{,}78$ mol

$n_{N_2} = 2 n_{O_2} = 2 \cdot 0{,}78$ mol = 1,56 mol

$PM_{N_2} = 28 \frac{g}{mol}$

$m_{N_2} = 43{,}68$ g.

$m_T = m_{O_2} + m_{N_2} = 25 + 43{,}68$ g. = 68,68 g

PV = nRT

PV = $(n_{O_2} + n_{N_2})$ RT

PV = $(n_{O_2} + n_{N_2})$ RT

PV = n_{O_2} RT → $\rho = \frac{\text{masa}}{\text{volumen}}$

R: 0,082 $\frac{atm\,L}{mol\,K}$

V = ~~300 L~~ 99,7

$\rho = \frac{68{,}68 g}{300 L} = 0{,}688$

PROBLEMA 9

NH_4Cl(s) → HCl(g) + NH_3(g)

400°C
673 K

pureza 80%
masa?

R = 70%

V = 65 L $P_T = P_{HCl} + P_{NH_3}$

$P_T = 0{,}95$ atm.

1 mol NH_4Cl → 1 mol HCl → 0,7 mol HCl (R)
1 mol NH_3 = 0,7 mol NH_3

$P_T V = n_T RT$

$\frac{P_T V}{RT} = n_T = 1{,}12$ mol → 0,56 mol.

0,7 mol HCl → 1 mol NH_4Cl
0,56 mol HCl → x = 0,8 mol NH_4Cl.

0,8 mol NH_4Cl → 80%
1 mol NH_4Cl = x ← 100%

$PM \cdot NH_4Cl = 35{,}5 + 4 + 14 = 53{,}5$.